Stability of LTI system

Stability is an important property that must be considered in any practical implementation of a system. We defined an arbitrary relaxed system as BIBO stable if and only if its output sequence $y(n)$ is bounded for every bounded input $x(n)$.

If $x(n)$ is bounded, there exists a constant M_x such that

$$|x(n)| \leq M_x < \infty$$

Similarly, if the output is bounded, there exists a constant M_y such that

$$|y(n)| \leq M_y < \infty$$

for all n.

Now, given such a bounded input sequence $x(n)$ to a linear time-invariant system, let us investigate the implications of the defination of stability on the characteristics of the system.

The convolution equation can be written as

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

Now, taking absolute value of both sides, we obtain

$$|y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right|$$

The absolute value of the sum of terms is always less than or equal to the sum of the absolute values of the terms. Hence

$$|y(n)| \leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)|$$

Here, $x(n-k)$ is delayed input signal. If input is bounded then its delayed version is also bounded i.e., if $|x(n)| \leq M_x$ then $|x(n-k)| \leq M_x$.

Putting this values we get,

$$|y(n)| \leq \sum_{k=-\infty}^{\infty} |h(k)|\, M_x$$

$$|y(n)| \leq M_x \sum_{k=-\infty}^{\infty} |h(k)|$$

We know that M_x is a finite number. We want output $y(n)$ to be bounded. That means $|y(n)|$ should be finite. So the output is bounded if the impulse response of the system satisfies the condition

$$S = \sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

That is, **a linear time-invariant system is stable if its impulse response is absolutely summable**. This condition is not only sufficient but it is also necessary to ensure the stability of the system.

To show that it is also a necessary condition, we must show that if $S = \infty$, then a bounded input can be found that will cause an unbounded output.
Such an input is the sequence with values

$$x(n) = \begin{cases} \dfrac{h^*(-n)}{|h(-n)|}, & h(n) \neq 0, \\ 0, & h(n) = 0, \end{cases}$$

Where $h^*(-n)$ is the complex conjugate of $h(n)$. The sequence $x(n)$ is clearly bounded by unity. However, the value of the output at $n = 0$ is

$$y(0) = \sum_{k=-\infty}^{\infty} x(-k)h(k) = \sum_{k=-\infty}^{\infty} \frac{|h(k)|^2}{|h(k)|} = S$$

Therefore, if $S = \infty$, it is possible for a bounded input sequence to produce an unbounded output sequence.

Example1: Determine the range of values of the parameter a for which the linear time-invariant system with impulse response $h(n) = a^n u(n)$ is stable.

Solution: The system is causal. Now

$$S = \sum_{k=-\infty}^{\infty} |h(k)| = \sum_{k=0}^{\infty} \left|a^k\right| = 1 + |a| + |a|^2 + \cdots$$

This geometric series converges to

$$\sum_{k=-\infty}^{\infty} \left|a^k\right| = \frac{1}{1 - |a|}$$

provided that $|a| < 1$, otherwise it diverges. Therefore, the system is stable if $|a| < 1$, otherwis it is unstable.

Example2: Determine the range of values of a and b for which the linear time invariant system with impulse response

$$h(n) = \begin{cases} a^n, n \geq 0 \\ b^n, n < 0 \end{cases}$$

is stable.
Solution: This system is non-casual.

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=-\infty}^{-1} |b|^n + \sum_{n=0}^{\infty} |a|^n$$

The second term converges for $|a| < 1$, the first term can be manipulate as

$$\sum_{n=-\infty}^{-1} |b|^n = \sum_{n=1}^{\infty} \frac{1}{|b|^n} = \frac{1}{|b|}[1 + \frac{1}{|b|} + \frac{1}{|b|^2} + \cdots]$$

$$= \beta(1 + \beta^2 + \cdots) = \frac{\beta}{1 - \beta}$$

where $\beta = \frac{1}{|b|}$ must be less than unity for the geometric series to converge. Cosnequently, the system is stable if both $|a| < 1 \; and \; |b| > 1$ are satisfied.

Causal Linear Time invariant systems

A causal system is whose output at time n depends only on present and past inputs but does not depends on future inputs. In other words, the output of the system at some time instant n, say $\mathrm{n} = n_0$, depends only on values of $x(n)$ for $n \leq n_0$.

In case of a linear time-invariant system, causality can be translated to a condition on the impuse response. To determine this relationship, let us consider a linear time-invariant system having an output at time $n = n_0$ given by the convolution formula

$$y(n_0) = \sum_{k=-\infty}^{\infty} h(k)x(n_0 - k)$$

Suppose that we subdivide the sum into two sets of terms, one set involving present and past values of the input [i.e., $x(n)$ for$n \leq n_0$] and one set involving future values of the input [i.e., $x(n)$ for $n > n_0$].

Thus we obtain

$$y(n_0) = \sum_{k=-\infty}^{-1} h(k)x(n_0 - k) + \sum_{k=0}^{\infty} h(k)x(n_0 - k)$$
$$= [h(-1)x(n_0 + 1) + h(-2)x(n_0 + 2) + \cdots]$$
$$+[h(0)x(n_0) + h(1)x(n_0 - 1) + h(2)x(n_0 - 2) + \cdots]$$

We observe that the terms in the second sum involve $x(n_0)$, $x(n_0 - 1), x(n_0 -$

Hence, **an LTI system is causal if and only if its impulse response is zero for negative values of n.**

Since for a causal system, $h(n) = 0$ for $n < 0$, the limits on the summation of the convolution formula may be modified to reflect this restriction. Thus we have the equivalent forms

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$$

$$= \sum_{k=-\infty}^{n} x(k)h(n-k)$$

Causality is requried in any real-time signal processing application, since at any given time n we have no access to future values of the input signal. Only the present and past values of the input signal are available in computing the present output.

If the input to a causal linear time-invariant system is a causal sequence (i.e., if $x(n) = 0$ for $n < 0$), the limits on the convolution formula are further restricted. In this case the two equivalent forms of the convolution formaula become

$$y(n) = \sum_{k=0}^{n} h(k)x(n-k)$$

$$= \sum_{k=0}^{n} x(k)h(n-k)$$

The response of a causal system to a causal input sequence is causal, since $y(n) = 0 \; for \; n < 0$.

To illustrate how the properties of LTI systems are reflected in the impulse response, let consider some linear and time invariant examples.

Ideal Delay

$$h(n) = \delta(n - n_d),$$

n_d a positive fixed integer

Moving Average

$$h(n) = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} \delta(n - k)$$

$$= \begin{cases} \dfrac{1}{M_1 + M_2 + 1}, & -M_1 \leq n \leq M_2 \\ 0, & otherwise \end{cases},$$

Accumulator

$$h(n) = \sum_{k=-\infty}^{n} \delta(k) = \begin{cases} 1, n \geq 0, \\ 0, n < 0, \end{cases} = u(n)$$

Forward Difference

$$h(n) = \delta(n+1) - \delta(n)$$

Backward Difference

$$h(n) = \delta(n) - \delta(n-1)$$

Given the impulse response of these basic systems, we can test the stablilty of each one by computing the sum

$$S = \sum_{n=-\infty}^{\infty} |h(n)|$$

For the ideal delay, moving average, forward difference, and backward difference, it is clear that $S < \infty$, since the impulse response has only a finite number of nonzero samples. Such systems are called finite-duration impulse response (FIR) systems. Clearly, FIR systems will be always be stable, as long as each of the impulse response values is finite in magnitude.

The accumulator, however, is unstable because

$$S = \sum_{n=0}^{\infty} u(n) = \infty$$

Also accumulator is unstable when a bounded input (the unit step) is given (see slide 87) it produces an unbounded output.
The impulse of the accumulator is infinite in duration. This is an example of the class of systems reffered to as infinite-duration impulse response (IIR) systems.

An example of an IIR system that is stable is a system whose impulse response is $h(n) = a^n u(n)$ with $|a| < 1$. In this case,

$$S = \sum_{n=0}^{\infty} |a|^n$$

If $|a| < 1$, the formula for the sum of the terms of an infinite geometric series gives

$$S = \frac{1}{1 - |a|} < \infty$$

If, on other hand, $|a| \geq 1$, the sum is infinite and the system is unstable.

Note:

1. The ideal delay $n_d \geq 0$ is causal and $n_d < 0$ is noncausal.
2. The accumulator and backward difference systems are causal
3. The forward difference is noncausal.
4. For the moving average, causality requries that $-M_1 \geq 0 \; and \; M_2 \geq 0$.

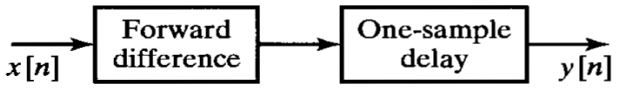
The concept of convolution as an operation between two sequences leads to the simplification of many problems involving systems.

A particular useful result can be stated for the ideal delay system. Since the output of the delay system is $y(n) = x(n - n_d)$, since the delay system has impulse response $h(n) = \delta(n - n_d)$, it follows that

$$x(n) * \delta(n - n_d) = \delta(n - n_d) * x(\mathrm{n}) = x(n - n_d)$$

That is, the convolution of a shifted impulse sequence with any signal $x(n)$ is easily evaluated by simplying shifting $x(n)$ by the displacement of the impulse.

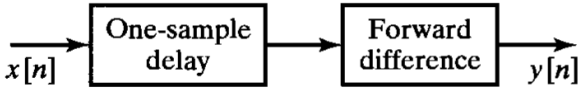
Consider the system which consists of a forward difference system cascaded with an ideal delay of one sample as in figure.



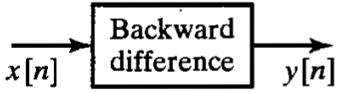
The impulse response of system is

$$h(n) = [\delta(n+1) - \delta(n)] * \delta(n-1)$$

According to cummulative property of convolution, the order in which system are cascaded does not matter, as long as they are linear and time invariant. Therefore we obtain same result when we delay the sequence first and the and compute forward difference as in figure.



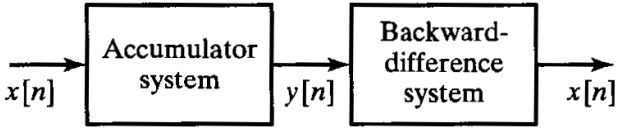
$$h(n) = \delta(n-1) * [\delta(n+1) - \delta(n)]$$
$$= \delta(n) - \delta(n-1)$$



Thus, $h(n)$ is identical to the impulse response of the backward difference system, i.e., ths cascade system can be repalced by a backward differnce system as in figure.

The noncausal forward difference systems have converted to causal systems by cascading them with a delay. In general, **any noncausal FIR system can be made causal by cascading it with a sufficient long delay**.

Consider another cascade system as in figure, a cascade combination of accumulator followed by a backward difference (or vice versa).



The impulse response of the cascade system is

$$h(n) = u(n) * [\delta(n) - \delta(n-1)]$$
$$= u(n) - u(n-1)$$
$$= \delta(n)$$

The cascade combination of an accumulator followed by a backward difference (or vice versa) yields a system whose overall impulse response is the impulse.

Since $x(n) * \delta(n) = x(n)$, thus the backward difference system compensates exactly for (or inverts) the effect of the accumulator.

The backward difference system is the inverse system for the accumulator and from commutative property of convolution, the accumulator is likewise the inverse system for the backward difference system.

In general, if a linear time-invariant system has impulse response $h(n)$, then its inverse system, if it exists, has impulse respone $h_i(n)$ defined by the relation

$$h(n) * h_i(n) = h_i(n) * h(n) = \delta(n)$$

Discrete-time system described by Difference Equations

The convolution summation formula

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h\,(n-k)$$

This equation suggests a means for the realization of the system. In the case of FIR systems, such realization involves additions, multiplications, and a finite number of memory location which is readily implemented directly.

If the system is IIR, however, its practical implementation as given by convolution is clearly impossible, since it requires an infinite number of memory locations, multiplications, and additions.

There is a practical and computationally efficient means for implementing a family of IIR systems. This family of discrete-time systems is more conveniently described by difference equations. This family of IIR systems is very useful in a variety of practical applications, including the implementation of digital filters, and the modeling of physical phenomena and physical systems.

Recursive and Nonrecursive Discrete-time Systems

If the response of any discrete time systems depends only on the input signals then the system is known as non-recursive discrete time system.

If we express the output of the system not only in terms of the present and past values of the input, but also in terms of the past output values, then that system is known as recursive system.

Example the cumulative average of a signal $x(n)$ in the interval $0 \leq k \leq n$ defined as,

$$y(n) = \frac{1}{n+1}\sum_{k=0}^{n} x(k)\,, \qquad n = 0,1,\ldots$$

This equation implies the computation of $y(n)$ requires the storage of all the input samples $x(k)$ for $0 \leq k \leq n$. Since n is increasing, our memory requirements grow linearly with time. However, $y(n)$ can be computed more efficiently by utilizing the previous output values $y(n-1)$. By a simple algebraic rearrangement, we obtain,

$$(n+1)y(n) = \sum_{k=0}^{n-1} x(k) + x(n) = ny(n-1) + x(n)$$

$$y(n) = \frac{n}{n+1} y(n-1) + \frac{1}{n+1} x(n)$$

Now the cumulative average y(n) can be computed recursively by multiplying the previous output value $y(n-1)$ by $n/(n+1)$, multiplying the present input $x(n)$ by $1/(n+1)$, and adding the two products as in figure. A system whose output $y(n)$ at time n depends on any number of past output values $y(n-1), y(n-2), \ldots$ is called recursive system.

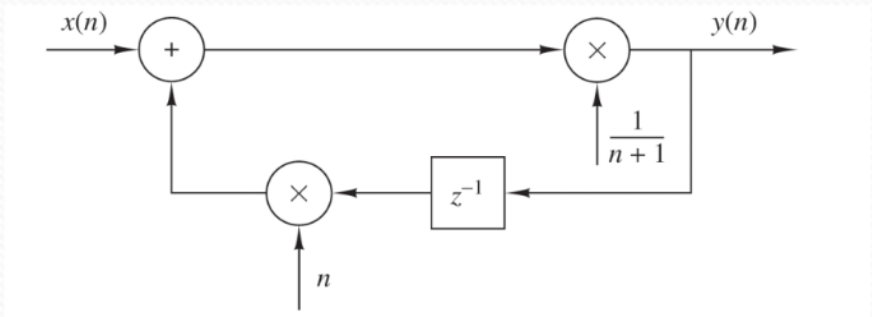
To determine the computation of recursive system, suppose we begin the process with $n = 0$ and proceed forward in time and we obtain

$y(0) = x(0)$

$y(1) = \frac{1}{2} y(0) + \frac{1}{2} x(0)$

$y(2) = \frac{2}{3} y(1) + \frac{1}{3} x(1)$

And so on.



Realization of a recursive cumulative averaging system.

1.9 Frequency Response of LTI systems

- Discrete time signals may be represented in a number of different ways.
- For example, sinusoidal and complex exponential sequences play a particularly important role in representing discrete time signals.
- This is because complex exponential sequences are eigenfunctions of linear time invariant systems and the response to a sinusoidal input is sinusoidal with the same frequency as the input an with amplitude and phase determined by the system.
- This fundamental property if linear time invariant systems makes representation of signals in terms of sinusoids or complex exponentials (i.e., Fourier representations) very useful in linear system theory.

Eigenfunctions for Linear Time-Invariant Systems

Consider an input sequence $x(n) = e^{j\omega n}\ for\ -\infty < n < \infty$, i.e., a complex exponential of radian frequency ω. The corresponding output of a linear time invariant system with impulse response $h(n)$ is

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)e^{j\omega(n-k)}$$

$$= e^{j\omega n} \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}$$

If we define

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}$$

Thus

$$y(n) = H(e^{j\omega})e^{j\omega n}$$

Consequently, $e^{j\omega n}$ is the eigenfunction of the system, and the associated eigenvalue is $H(e^{j\omega})$.

$H(e^{j\omega})$ describes the change in complex amplitude of a complex exponential input signal as a function of the frequency ω.

The eigenvalue $H(e^{j\omega})$ is called the frequency response of the system.

In general, $H(e^{j\omega})$ is complex and can be expressed in terms of its real and imaginary parts as

$$H\left(e^{j\omega}\right) = H_R\left(e^{j\omega}\right) + \mathrm{j}H_I(e^{j\omega})$$

Or in terms of magnitude and phase as

$$H\left(e^{j\omega}\right) = \left|H(e^{j\omega})\right| e^{j\measuredangle H(e^{j\omega})}$$

Example: Frequency Response of the Ideal Delay System

Consider the ideal delay system defined by

$$y(n) = x(n - n_d)$$

Where n_d is a fixed integer. If we consider $x(n) = e^{j\omega n}$ as input to this system, then, we have

$$y(n) = e^{j\omega(n-n_d)} = e^{j\omega n} e^{-j\omega n_d}$$

The frequency response of the ideal delay is therefore

$$H\left(e^{j\omega}\right) = e^{-j\omega n_d}$$

The real an imaginary parts of the frequency response are

$$H_R\left(e^{j\omega}\right) = \cos(\omega n_d)$$

$$H_I\left(e^{j\omega}\right) = -\sin(\omega n_d)$$

The magintude and phase are

$$\left|H(e^{j\omega})\right| = 1,$$

$$\sphericalangle H\left(e^{j\omega}\right) = -\omega n_d$$

Example 2: Sinusoidal Response of LTI Systems

Since it is simple to express a sinusoid as a linear combination of complex expoenetials, let us consider a sinusoidal input

$$x(n) = A\cos(\omega_0 n + \phi) = \frac{A}{2}e^{j\phi}e^{j\omega_0 n} + \frac{A}{2}e^{-j\phi}e^{-j\omega_0 n}$$

Thus splitting the response as $x_1(n)$ and $x_2(n)$ as

$x_1(n) = \frac{A}{2}e^{j\phi}e^{j\omega_0 n}$ and $x_2(n) = \frac{A}{2}e^{-j\phi}e^{-j\omega_0 n}$ the response are

$$y_1(n) = H\left(e^{j\omega_0}\right)\frac{A}{2}e^{j\phi}e^{j\omega_0 n}$$

$$y_2(n) = H\left(e^{-j\omega_0}\right)\frac{A}{2}e^{-j\phi}e^{-j\omega_0 n}$$

Thus, the total response is

$$y(n) = \frac{A}{2}\left[H\left(e^{j\omega_0}\right)e^{j\phi}e^{j\omega_0 n} + H\left(e^{-j\omega_0}\right)e^{-j\phi}e^{-j\omega_0 n}\right]$$

If $h(n)$ is real, it can be shown that $H\left(e^{-j\omega_0}\right) = H^*\left(e^{j\omega_0}\right)$ (See slide 43)
Hence,

$$y(n) = A\left|H\left(e^{j\omega_0}\right)\right| \cos(\omega_0 n + \phi + \theta)$$

Where $\theta = \sphericalangle H\left(e^{j\omega_0}\right)$ is the phase of the system function at frequency ω_0.

Let use ideal delay with

$$\left|H(e^{j\omega})\right| = 1, \qquad \sphericalangle H\left(e^{j\omega}\right) = -\omega n_d$$

therefore,

$$y(n) = A\cos(\omega_0 n + \phi - \omega n_d) = A\cos[\omega_0(n - n_d) + \phi]$$

Which is consistent with what we would obtain directly usnig the definition of the ideal delay system.

The frequency response of discrete time linear time invariant system is always a periodic function of the frequency variable ω with period 2π.

To prove, we have

$$H\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} h(k)e^{-j\omega n}$$

Let substitute $\omega\ by\ \omega + 2\pi$ we get

$$H\left(e^{j(\omega+2\pi)}\right) = \sum_{n=-\infty}^{\infty} h(k)e^{-j(\omega+2\pi)n}$$

Using the fact that $e^{\pm j(2\pi)n} = 1$

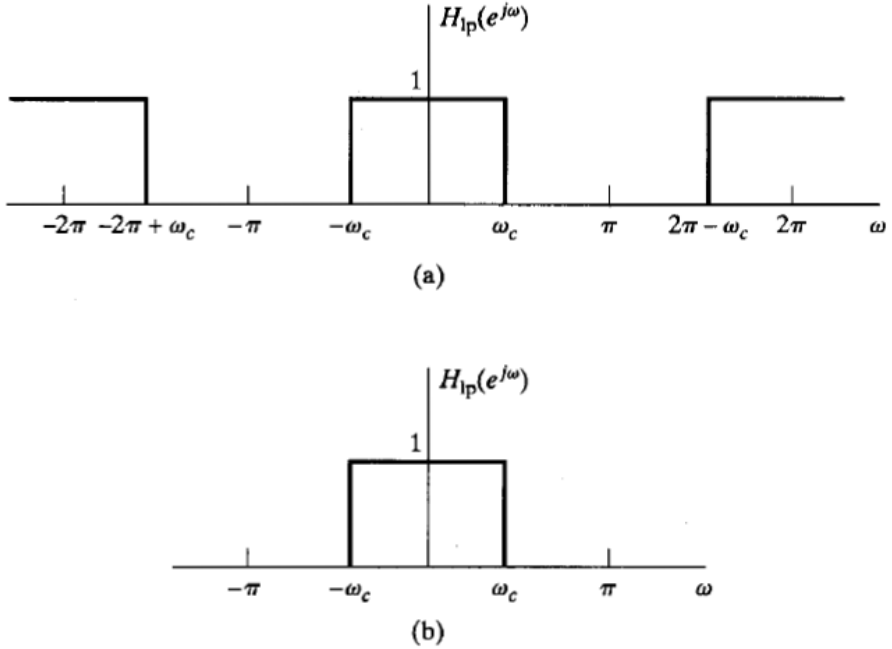
Therefore

$$H\left(e^{j(\omega+2\pi)}\right) = H\left(e^{j\omega}\right)$$

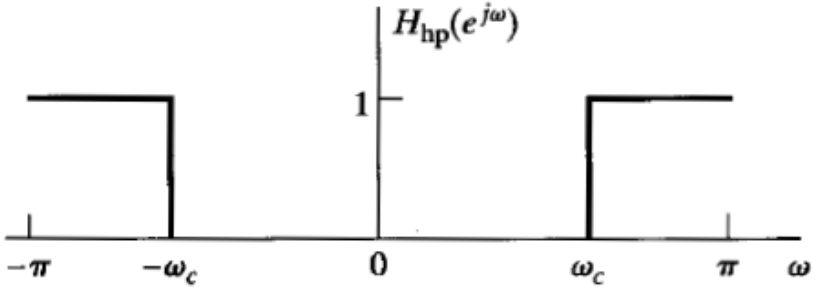
In general

$$H\left(e^{j(\omega+2\pi r)}\right) = H\left(e^{j\omega}\right), for\ r\ an\ integer$$

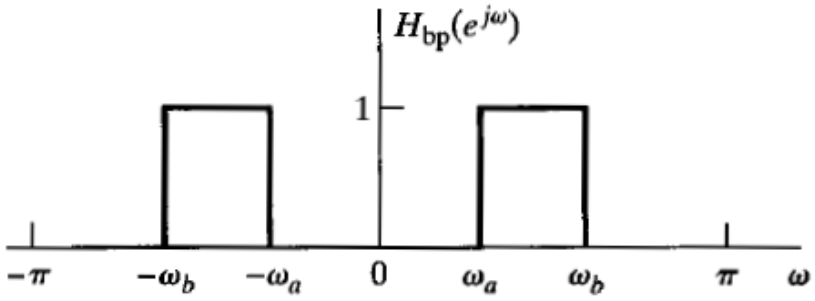
Since $H(e^{j\omega})$ is periodic with period 2π, and since the frequencies ω and $\omega + 2\pi$ are indistinguishable, it follows that we need only specify $H(e^{j\omega})$ over an interval of length 2π , e.g., $0 \leq \omega \leq 2\pi$ or $-\pi \leq \omega \leq \pi$.



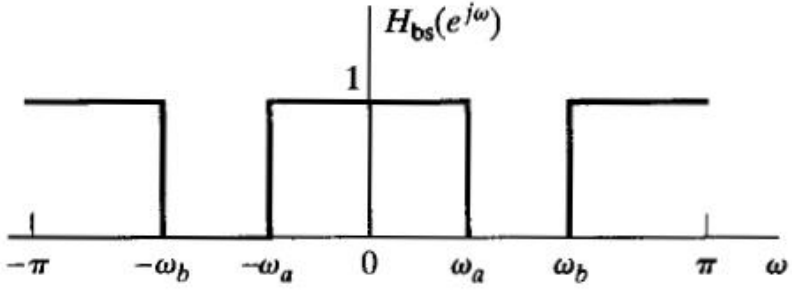
Ideal lowpass filter showing (a) periodicity of the frequency response and (b) one period of the periodic frequency response.



Highpass filter.



Bandpass filter.



Bandstop filter.

In each case, the frequency response is periodic with period 2π. Only one period is shown.

Frequency Response of the Moving Average System

The impulse response of the moving average system is

$$h(n) = \begin{cases} \dfrac{1}{M_1 + M_2 + 1}, & -M_1 \le n \le M_2, \\ 0, & otherwise \end{cases}$$

The frequency response is

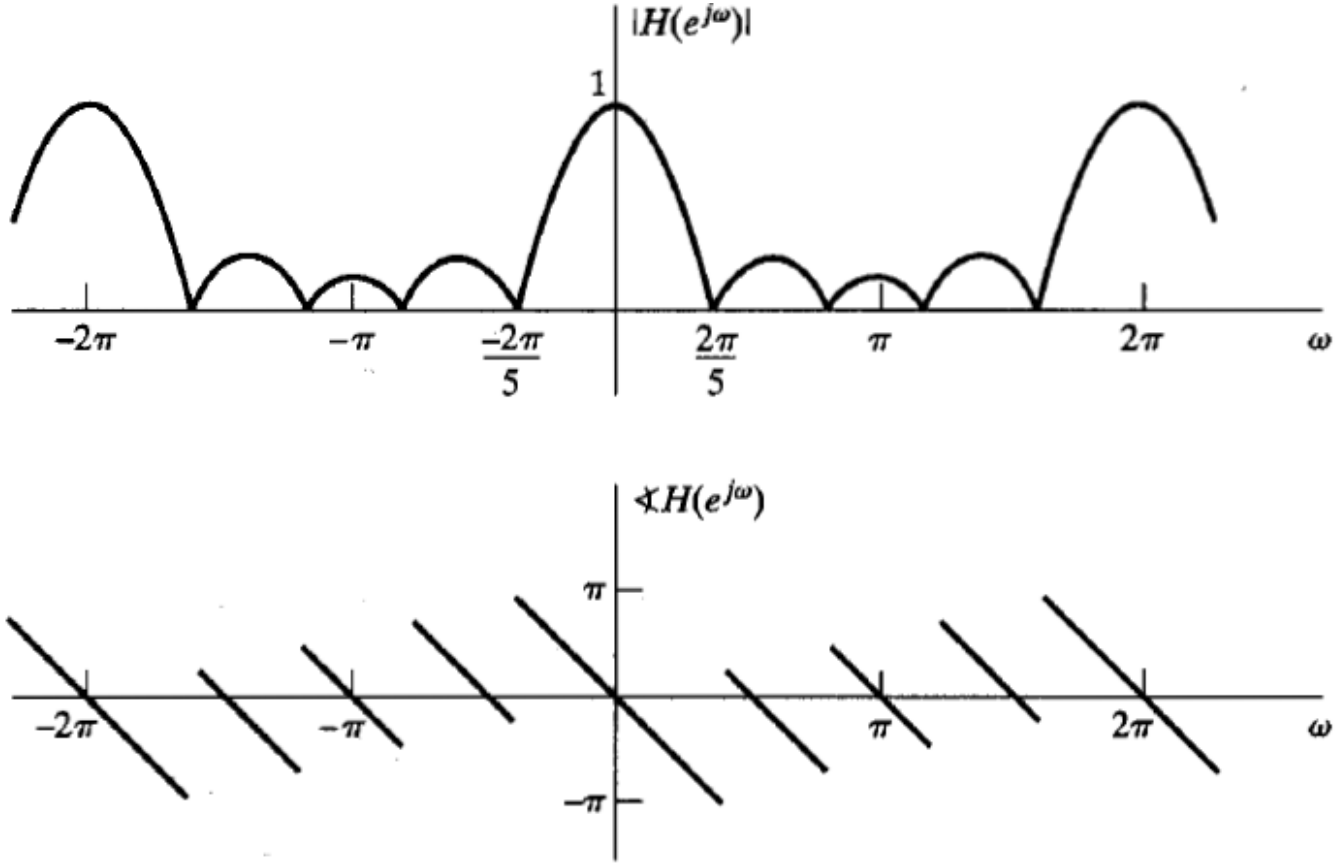
$$H\left(e^{j\omega}\right) = \frac{1}{M_1 + M_2 + 1} \sum_{n=-M_1}^{M_2} e^{-j\omega n}$$

$$H(e^{j\omega}) = \frac{1}{M_1 + M_2 + 1} \frac{e^{j\omega M_1} - e^{-j\omega(M_2+1)}}{1 - e^{-j\omega}}$$

$$= \frac{1}{M_1 + M_2 + 1} \frac{e^{j\omega(M_1+M_2+1)/2} - e^{-j\omega(M_1+M_2+1)/2}}{1 - e^{-j\omega}} e^{-j\omega(M_2-M_1+1)/2}$$

$$= \frac{1}{M_1 + M_2 + 1} \frac{e^{j\omega(M_1+M_2+1)/2} - e^{-j\omega(M_1+M_2+1)/2}}{e^{j\omega/2} - e^{-j\omega/2}} e^{-j\omega(M_2-M_1)/2}$$

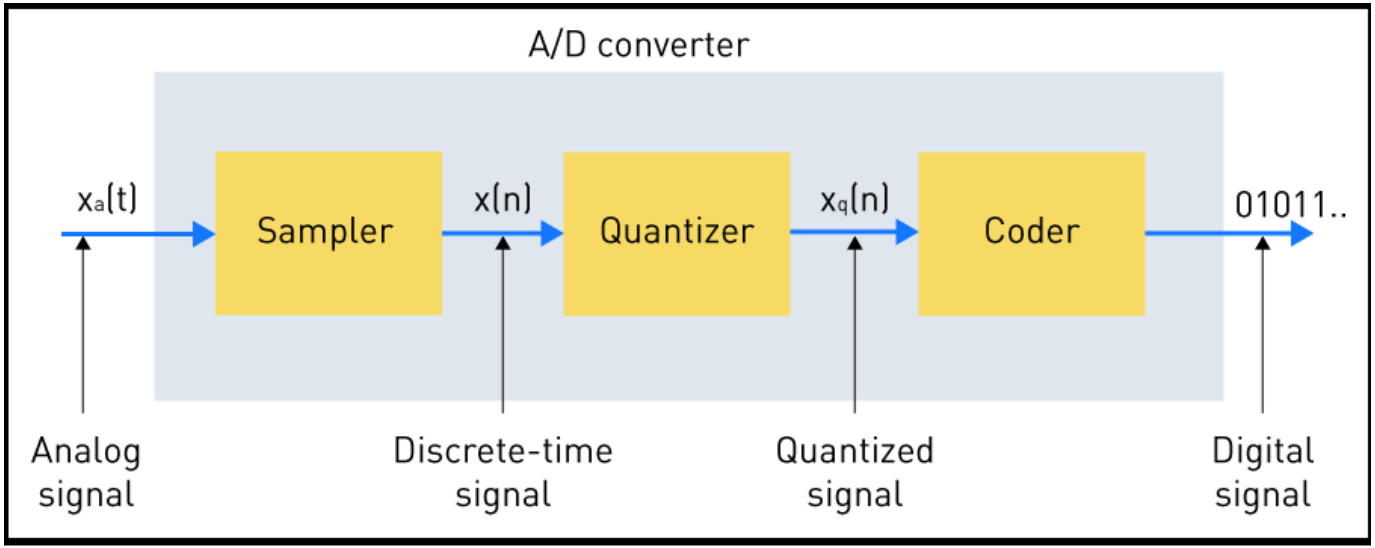
$$= \frac{1}{M_1 + M_2 + 1} \frac{\sin[\omega(M_1 + M_2 + 1)/2]}{\sin(\omega/2)} e^{-j\omega(M_2-M_1)/2}.$$



(a) Magnitude and (b) phase of the frequency response of the moving-average system for the case $M_1 = 0$ and $M_2 = 4$.

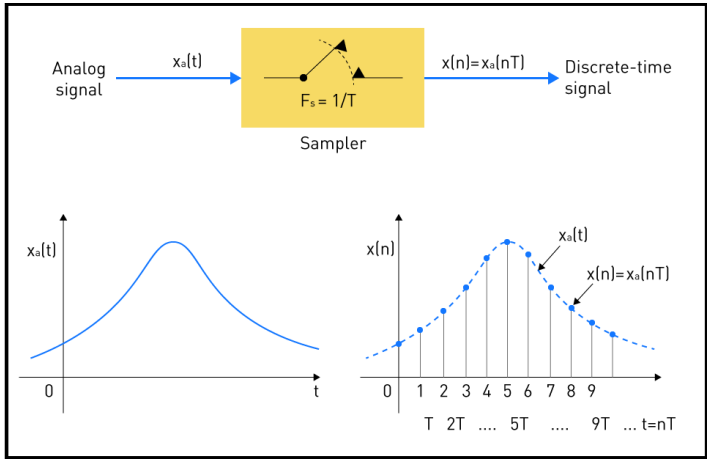
1.10 Sampling continuous signals and spectral properties of sampled signals

- Most of the signals of practical interest are analog in nature, such as voice signal, biological signal, communication signals, etc.
- To process analog signal by digital means, it is first necessary to convert them into digital form. This procedure is known as analog to digital conversion.
- Analog to digital conversion is a three step process as shown in the block diagram.



a. Sampling

- ✓ A continuous analog signal is first transformed into a discrete digital signal by sampling.
- ✓ The procedure entails periodic measurement or snapshots of the analog signal's amplitude at predetermined periods in time.
- ✓ The discrete set of data points that picture, or samples, reflect the original signal in the digital realm.
- ✓ The sampling rate or sampling frequency is the frequency at which these photographs or samples are taken.



In mathematical terms, if x(t) is a continuous-time analog signal, the sampled signal x(n) can be represented as

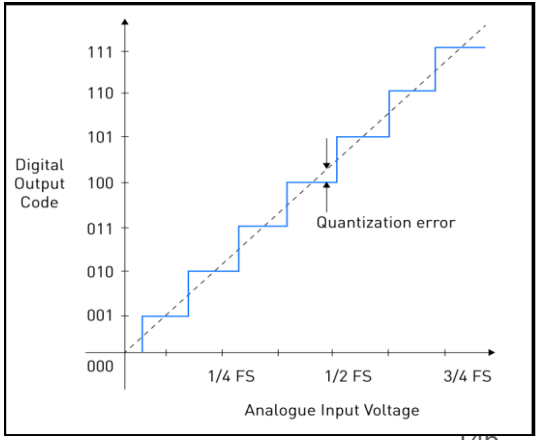
$x(n) = x(nT_s)$

Where T_s is the sampling period (the time period between the consecutive samples), n is an integer and x(n) denotes the value of the signal at the nth sample.

Sample and hold circuit are typically used to execute the sampling process. These circuit collects and hold the signal value for a brief period of time.

b. Quantization

- ✓ Quantization comes after sampling as a crucial step in converting continuous analog signals to digital signals.
- ✓ A continuous set of values is quantized into a discrete set of values.
- ✓ During the analog to digital conversion process, each sampled value is matched with the closest value among a limited number of discrete levels.
- ✓ Since quantization converts a continuous collection of values to a discrete set, it naturally involves an approximation inaccuracy.
- ✓ The discrepancy between the actual sampled value and the quantized value to which it is mapped is referred to as quantization error.
- ✓ Since the nature of quantization error is largely unpredictable, it may be though of as noise added to the signal.
- ✓ It can, however, be examined and its consequences recognized.
- ✓ The error is often constrained to $\pm\,{}^{1}/_{2}$ of the quantization step size, which is the separation between neighboring quantization levels.



Quantization Levels and Resolution

- ✓ The resolution of the quantizer is measured in bits that determines the number of quantization levels.
- ✓ The number of quantization levels increases with resolution, while the quantization error decreases.
- ✓ There are 2^N quantization levels for an N-bit quantizer. An 8-bit quantizer will have 256 quantization levels.
- ✓ The signal range divided by quantization levels yields the quantization step size. For example, the step size would be 10/256 = 0.039 volts for a signal that range from 0 to 10 volts and has 256 quantization levels.
- ✓ For high-fidelity applications, high resolution quantization is preferred because it lowers quantization error.
- ✓ It also needs extra bit for representation, which results in a trade off in terms of bandwidth and storage.

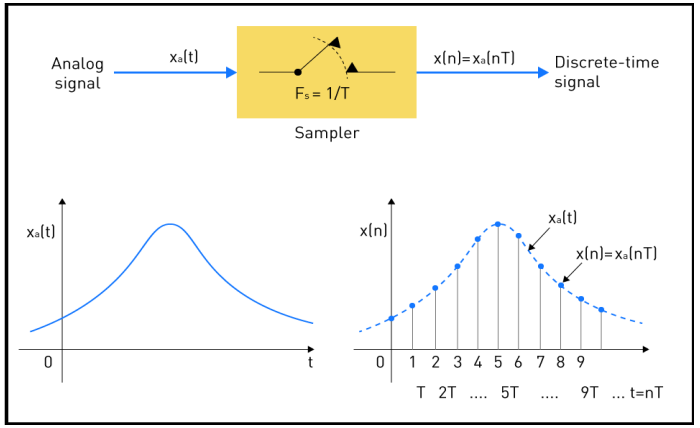
c. Encoding

- ✓ The process of converting an analog signal to a digital signal continues with encoding after the analog signal has been sampled and quantized.
- ✓ Each quantized value is represented using a binary code.
- ✓ A distinct binary code is assigned to each quantized samples.

Digital to Analog Conversion

- ➢ In many cases of practical interest it is desirable to convert the processed digital signals into analog form.
- ➢ The process of converting a digital signal to analog signal is known as digital to analog (D/A) conversion.
- ➢ Sampling does not result in a loss of information, nor does it introduce distortion in the signal if the signal bandwidth is finite.
- ➢ In principle, the analog signal can be reconstructed from the samples, provided that the sampling rate is sufficient high to avoid the problem commonly called aliasing.
- ➢ On the other hand, quantization is a noninvertible or irreversible process that results in signal distortion.

Sampling of Analog Signals



- There are many ways to sample an analog signal.
- The most often used sampling in practice is periodic or uniform sampling.
- The mathematical relationship is given as

$$x(n) = x_a(nT) \quad -\infty < n < \infty$$

Where x(n) is the discrete time signal obtained by taking samples of analog signal $x_a(t)$ every T seconds. This procedure is as in figure.

- The time interval T between successive samples is called the sampling period or sample interval and $^1/_T = F_s$ is called the sampling rate (samples per second) or the sampling frequency (hertz).
- Periodic samples establishes a relationship between the time variables t and n of continuous time and discrete time signals, respectively as

$$t = nT = \frac{n}{F_s}$$

- There exists a relationship between the frequency variable F (or Ω) for analog signals and the frequency variable f (or ω) for discrete time signal.

Consider an analog sinusoidal signal of the form

$$x_a(t) = A\cos(2\pi Ft + \theta)$$

Which, when sampled periodically at the rate $F_s = {}^{1}/_{T}$ samples per second, yields

$$x_a(nT) \equiv x_a(t) = A\cos(2\pi FnT + \theta)$$

$$= A\cos(\frac{2\pi Fn}{F_s} + \theta)$$

A discrete time sinusoidal signal is expressed as

$$x(n) = A\cos(\omega n + \theta)\,, -\infty < n < \infty$$

$$= A\cos(2\pi fn + \theta)$$

Comparing we get that the frequency variables F and f are linearly related as

$$f = \frac{F}{F_s}\ or\ \omega = \Omega T$$

The range of frequency variable F (or Ω) for continuous time sinusoids are

$$-\infty < F \text{ (or } \Omega) < \infty$$

And for discrete time sinusoids $-\frac{1}{2} < f < \frac{1}{2}$ or $-\pi < \omega < \pi$

The frequency of the continuous time sinusoid when sampled at a rate $F_s = {}^1/_T$ must fall in the range

$$-\frac{1}{2T} = -\frac{F_s}{2} \leq F \leq \frac{F_s}{2} = \frac{1}{2T}$$

Or equivalently,

$$-\frac{\pi}{T} = -\pi F_s \leq \Omega \leq \pi F_s = \frac{\pi}{T}$$

Continuous time frequencies	Ω, F
Discrete time frequencies	ω, f
Sampling frequency	F_s or $\frac{1}{T}$
Conversion relations	$\omega = \frac{\Omega}{F_s} = \Omega T$
	$f = \frac{F}{F_s} = FT$
Range of continuous time frequencies	$-\infty < \Omega < \infty$
	$-\infty < F < \infty$
Range of discrete time frequencies	$-\frac{1}{2} \leq f \leq \frac{1}{2}$
	$-\pi \leq \omega \leq \pi$
	$-\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$
	$-\pi F_s \leq F \leq \pi F_s$

From these relations the fundamental difference between continuous time and discrete time signals is in their range of values of the frequency variables.

Periodic sampling of a continuous time signal implies a mapping of the infinite frequency range for the variable F (or Ω) into a finite frequency range for the variable f (or ω).

Since the highest frequency in a discrete time signal is $\omega = \pi$ of $f = \frac{1}{2}$, it follows that, with a sampling rate F_s, the corresponding highest value of F and Ω are

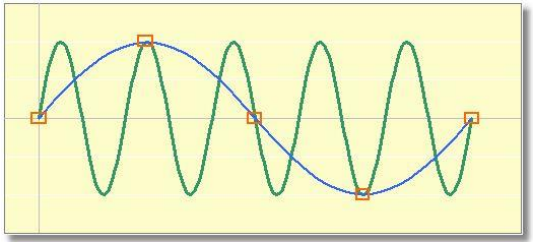
$$F_{max} = \frac{F_s}{2} = \frac{1}{2T}$$

$$\Omega_{max} = \pi F_s = \frac{\pi}{T}$$

Consider two analog sinusoidal signal $x_1(t) = \cos 2\pi(10)t$ and $x_2(t) = \cos 2\pi(50)t$ sampled at a rate $F_s = 40Hz$. The corresponding discrete time signals or sequences are

$$x_1(n) = \cos 2\pi(\frac{10}{40})\, n = \cos\frac{\pi}{2}\, n$$

$$x_2(n) = \cos 2\pi(\frac{50}{40})\, n = \cos\frac{5\pi}{2}\, n$$



However $\cos\frac{5\pi}{2}\, n = \cos(2\pi n + \frac{\pi}{2}\, n) = \cos\frac{\pi}{2}\, n$. Hence $x_1(n) = x_2(n)$.

Thus the sinusoidal signals are identical and, consequently, indistinguishable. The sampled value generated by $\cos\frac{\pi}{2}\, n$, there is some ambiguity as to whether these sampled values correspond to $x_1(t)$ or $x_2(t)$. Since $x_2(t)$ yields exactly the same value as $x_1(t)$ when the two are sampled at $F_s = 40Hz$.

The frequency $F_2 = 50Hz$ is an alias of the frequency $F_1 = 10Hz$ at the sampling rate of $40Hz$.

All of the sinusoids $\cos 2\pi(F_1 + 40k)t, k = 1, 2, 3, 4$ ... sampled at 40 samples per second yield identical values. Consequently, they are all aliases of $F_1 = 10Hz$.

In general, the sampling of a continuous time sinusoidal signal

$$x_a(t) = A\cos(2\pi F_0 t + \theta)$$

With a sampling rate $F_s = {}^{1}/_{T}$ results in a discrete time signal

$$x(n) = A\cos(2\pi f_0 n + \theta)$$

where $f_0 = {}^{F_0}/_{F_s}$, is the relative frequency of the sinusoid.

If we assume that $-\frac{F_s}{2} \leq F_0 \leq \frac{F_s}{2}$ the frequency f_0 of $x(n)$ is in the range $-\frac{1}{2} \leq f_0 \leq \frac{1}{2}$, which is the frequency range for discrete time signal.

In this case, the relationship between F_0 and f_0 is one-to-one, and hence it is possible to identify (or reconstruct) the analog signal $x_a(t)$ from the samples x(n).

If the sinusoids $x_a(t) = A\cos(2\pi F_k t + \theta)$ where $F_k = F_0 + kF_s,\ \mathrm{k} = \pm 1, \pm 2$ ...

are sampled at a rate F_s, it is clear that the frequency F_k is outside the fundamental frequency range $-\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$. Consequently the sampled signal is

$$x_a(nT) \equiv x_a(t) = A\cos\left(2\pi\frac{F_0 + kF_s}{F_s}n + \theta\right) = A\cos\left(2\pi\frac{F_0}{F_s}n + \theta + 2\pi kn\right) = A\cos(2\pi f_0 n + \theta)$$

Thus an infinite number of continuous time sinusoids is represented by sample the same discrete time signal.

Consequently, if we are given the sequence x(n), an ambiguity exists as to which continuous time signal $x_a(t)$ these value represents.

The frequency $F_k = F_0 + kF_s, -\infty < k < \infty \ (k\ integer)$ are indistinguishable from the frequency F_0 after sampling and hence they are aliases of F_0.

Since ${}^{F_s}/_{2}$, which corresponds to $\omega = \pi$, is the highest frequency that can be represented uniquely with a sampling rate F_s, it is a simple matter to determine the mapping of any (alias) frequency above ${}^{F_s}/_{2}$ or $\omega = \pi$ as the pivotal point and reflect or "fold" the alias frequency to the range $0 \leq \omega \leq \pi$.

Since the point of reflection is ${}^{F_s}/_{2}$ or $\omega = \pi$, the frequency ${}^{F_s}/_{2}$ or $\omega = \pi$ is called the folding frequency.

Example :Consider the analog signal $x_a(t) = 3\cos(100\pi t)$

i. Determine the minimum sampling rate required to avoid aliasing.

ii. If the signal is sampled at the rate $F_s = 200\ Hz$. What is the discrete-time signal obtain after sampling?

iii. What is the frequency $0 < F < {}^{F_s}/_{2}$ of the a sinusoid that yields samples identical to those obtained for $F_s = 200\ Hz$.

iv. Repeat (ii) and (iii) if the signal is sampled at the rate $F_s = 75\ Hz$.

Solution :

i. The frequency of the analog signal is $F = 50Hz$. Hence the minimum sampling rate required to avoid aliasing is $F_s = 100\ Hz$.

ii. If the signal is sampled at $F_s = 200\ Hz$, the discrete time signal is $x(n) = 3\cos\frac{100\pi}{200}n = 3\cos\frac{\pi}{2}n$

iii. For the sampling rate of $F_s = 200Hz$, we have $f_0 = {}^{F_0}/_{F_s}$, thus $F_0 = f_0 F_s = 200 f_0$ from (ii) $f_0 = \frac{1}{4}$ thus $F_0 = 50Hz$,and $y_a(t) = 3\cos(2\pi F_0 t) = 3\cos(100\pi t)$

iv. If the signal is sampled at $F_s = 75\ Hz$, the discrete time signal is

$$x(n) = 3\cos\frac{100\pi}{75}n = 3\cos\frac{4\pi}{3}n = 3\cos\left(2\pi - \frac{2\pi}{3}\right)n = 3\cos\frac{2\pi}{3}n$$

For the sampling rate of $F_s = 75Hz$, we have $f_0 = {}^{F_0}/_{F_s}$, thus $F_0 = f_0 F_s = 75 f_0$

from (iv) $f_0 = \frac{1}{3}$ thus F_0 = 25Hz ,and $y_a(t) = 3\cos(2\pi F_0 t) = 3\cos(50\pi t)$

Sampled at $F_s = 75Hz$ identical samples. Hence F=50Hz is an alias of F=25Hz for the sampling rate $F_s = 75Hz$.

Note: To convert to discrete signal, use analog frequency F_0 and sampling frequency F_s and to convert back to analog signal from digital use digital frequency f_0 with sampling frequency F_s. Use the formula $f_0 = {}^{F_0}/_{F_s}$.

The Sampling Theorem

If the highest frequency contained in an analog signal $x_a(t)$ is $F_{max} = B$ and the signal is sampled at a rate $F_s > 2F_{max} \equiv 2B$, then $x_a(t)$ can be exactly recovered from its samples values using the interpolation function.

The sampling rate $F_N = 2B = 2F_{max}$ is called Nyquist rate. This is the boundary condition value required to avoid aliasing.

The sampling theorem states that a continuous time signal can be completely represented in its samples and recover that into an original form if the sampling frequency is greater than or equal to twice of the maximum frequency. In this situation aliasing does not occur.

Nyquist rate $F_N = 2F_{max} = F_s$

The folding frequency = $\frac{F_s}{2}$

Resolution $\Delta = \frac{X_{max} - X_{min}}{L-1}$

where X_{max} = $A_1 + A_2$ and X_{min} = $-X_{max}$ L = no. of levels

A_1 *and* A_2 are amplitude of signal.

Quantization

- ✓ Process of conveting a discrete time continuous amplitude signal into a digital signal

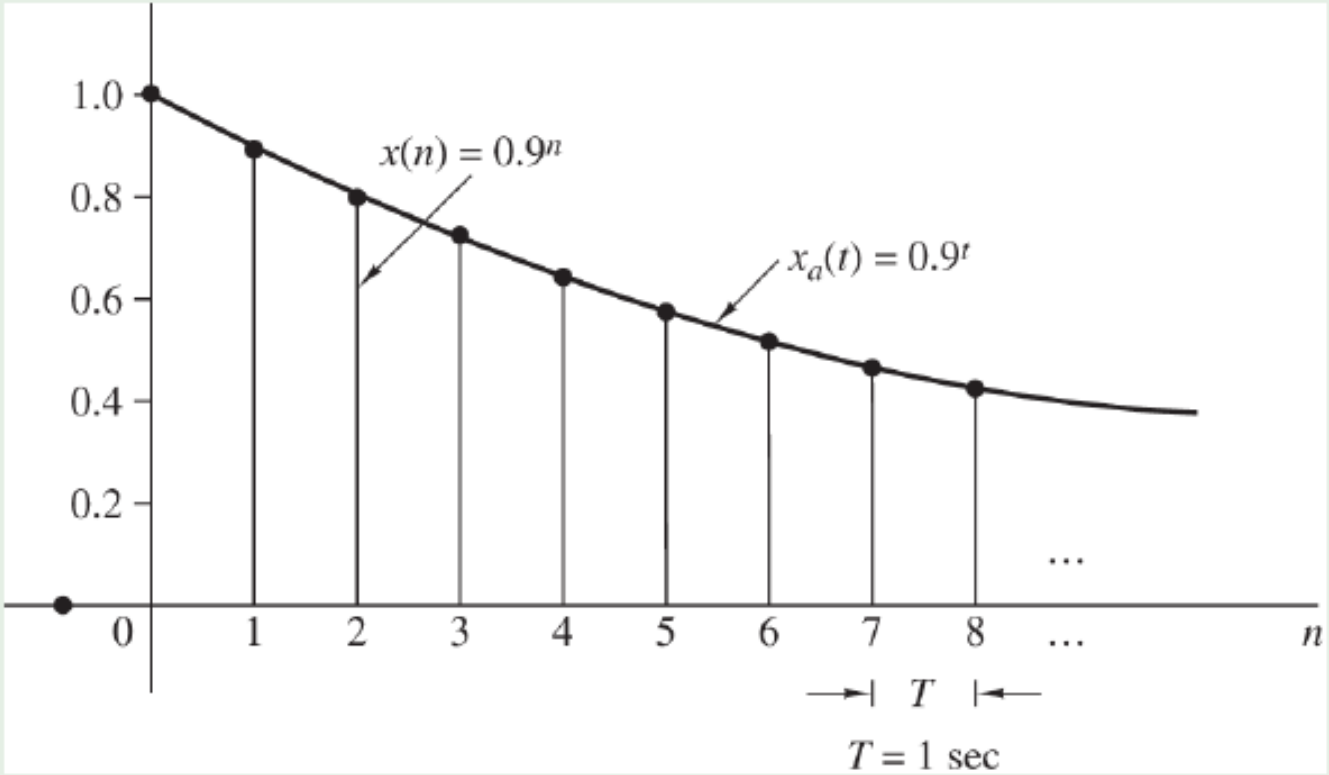
$$x_q(n) = Q[x(n)] = sequence\ of\ quantized\ samples$$

- ✓ Each sample value is expressed as a finite number of digits
- ✓ Error introduced is called quantization error or quantization noise.

$$e_q(n) = x_q(n) - x(n)$$

Example: Consider discrete time signal

$$x(n) = \begin{cases} 0.9^n, n \geq 0 \\ 0, n < 0 \end{cases} \ obtained\ by\ sampling\ x_a(t) = 0.9^t, t \geq 0\ with\ F_s = 1Hz$$



- Following table shows values of first 10 samples of $x(n)$
  - Description of sample value $x(n)$ requires n significant digits

n	$x(n)$ Discrete-time signal	$x_q(n)$ (Truncation)	$x_q(n)$ (Rounding)	$e_q(n) = x_q(n) - x(n)$ (Rounding)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	-0.01
3	0.729	0.7	0.7	-0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	-0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	-0.03046721
9	0.387420489	0.3	0.4	0.012579511

- Assume using one significant digit. To eliminate excess digits
  - do **truncation**
  - or do **rounding**

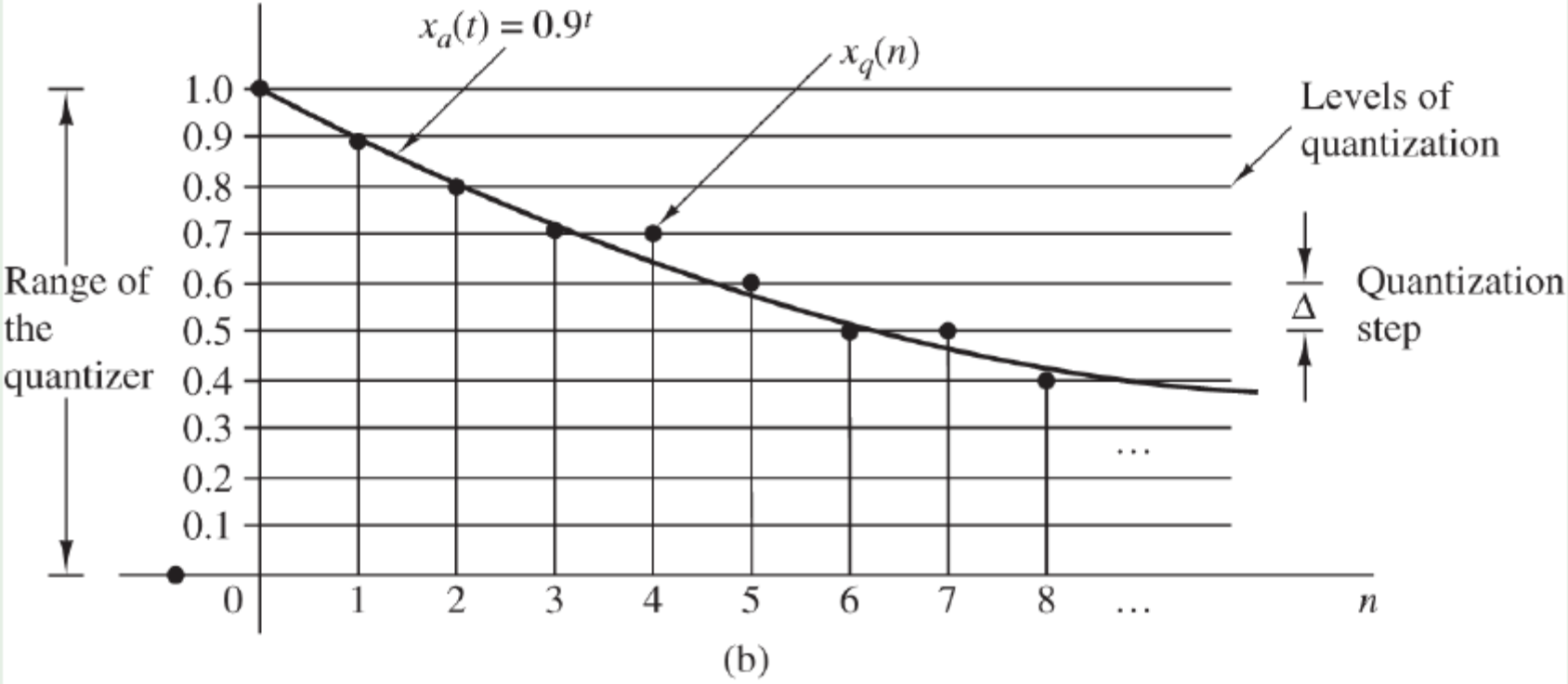


Figure 1.4.7 Illustration of quantization.

- Quantization step size (or resolution) $= \Delta = \frac{1-0}{11-1} = 0.1$

- Range of quantization error $e_q(n)$ in rounding

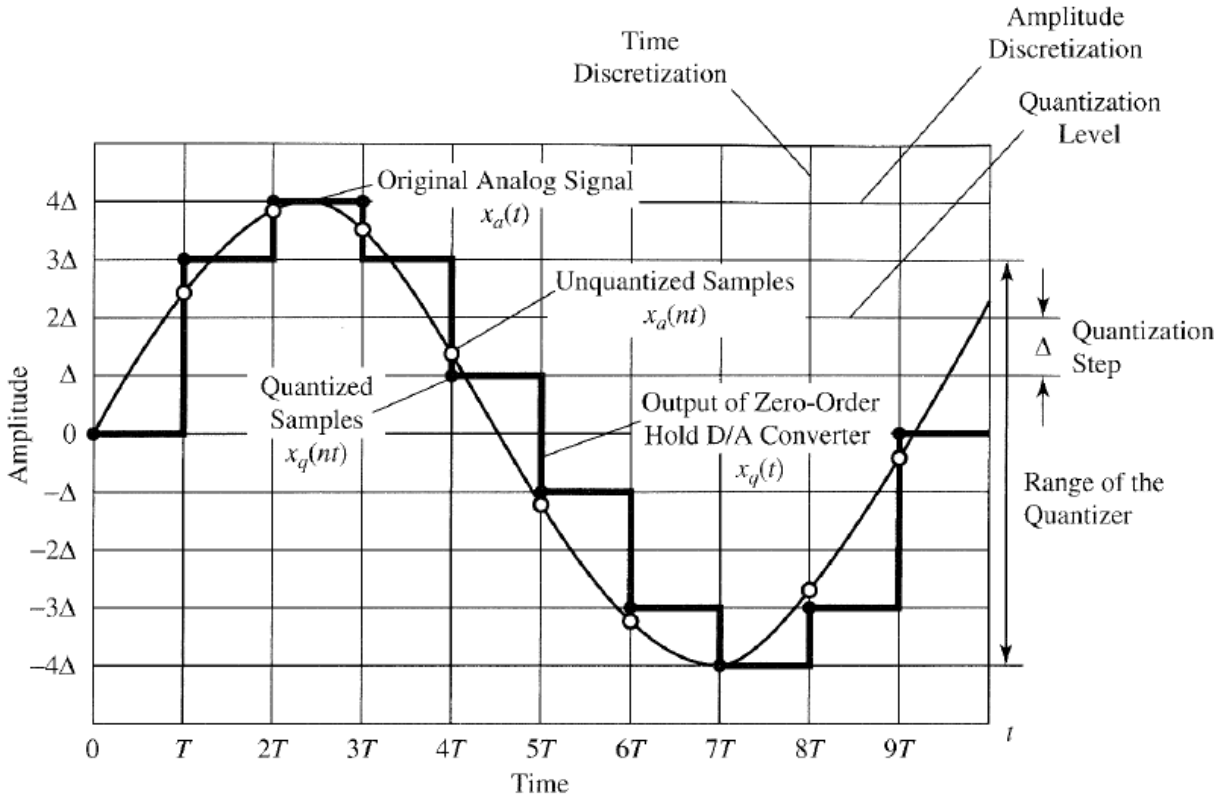
$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

$$\Delta = \frac{x_{max} - x_{min}}{L - 1}$$

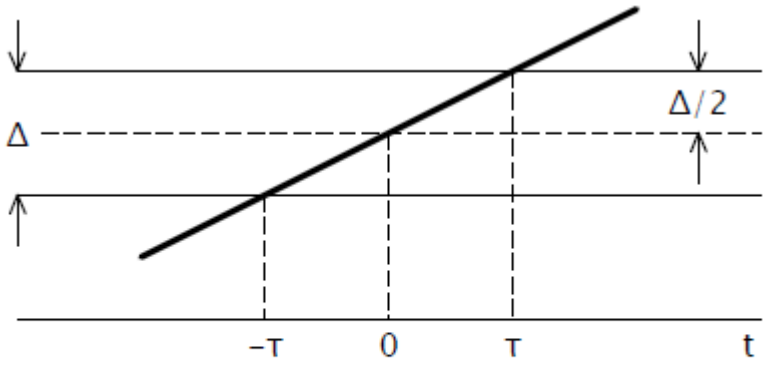
  - x_{min} and x_{max} = minimum and maximum value of $x(n)$
  - L = number of quantization levels
  - Dynamic range of signal = $x_{max} - x_{min}$
- Quantization of analog signals always results in a loss of information

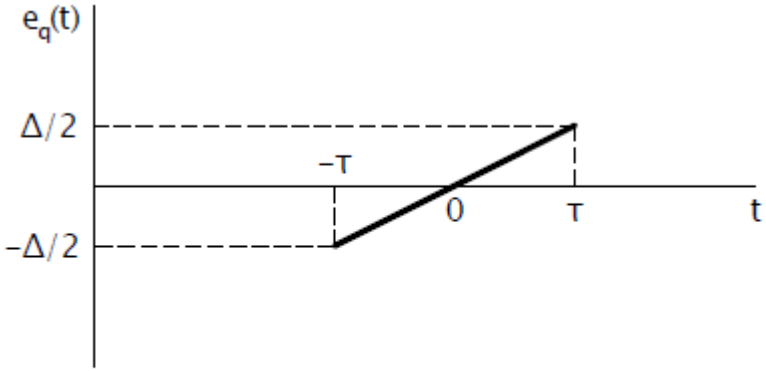
Quantization of Sinusoidal Signals

- Sampling and quantization of $x_a(t) = A\cos\Omega_0 t$



- If F_s satisfies sampling theorem, quantization is the only error in A/D process
  - Thus we can evaluate quantization error by quantizing $x_a(t)$ instead of $x(n) = x_a(nT)$
- $x_a(t)$ is almost linear between quantization levels
  - Quantization error $= e_q(t) = x_a(t) - x_q(t)$





  - τ denotes time that $x_a(t)$ stays within quantization levels

Example: 1. Consider the analog signal $x_a(t) = 3\cos 50\pi t + 10\sin 300\pi t + \cos 100\pi t$

Solution: The frequency present in the signal above are $F_1 = 25Hz$, $F_2 = 150Hz$, and $F_3 = 50Hz$. Thus $F_{max} = 150Hz$ and the Nyquist rate is $F_N = 2F_{max} = 300Hz$.

Note: The signal component $10\sin 300\pi t$, sampled at the Nyquist rate $F_N = 300Hz$, results in the samples $10\sin \pi n$, which is identically zero. In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.

This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have $10\sin(300\pi t + \theta)$ sampled at the Nyquist rate $F_N = 300Hz$ which yields the samples

$$10\sin(\pi n + \theta) = 10(\sin \pi n \cos\theta + \cos \pi n \sin\theta) = 10\cos \pi n \sin\theta = (-1)^n 10\sin\theta$$

Thus if $\theta \neq 0\ or\ \pi$, the samples of the sinusoid taken at the Nyquist rate are not all zero. However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown. A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a higher rate than Nyquist rate.

2. Consider the analog signal $x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12000\pi t$

i) What is the Nyquist rate for this signal?

ii) Assume now that we sample this signal using a sampling rate $F_s = 5000$ samples/s. what is the discrete time signal obtained after sampling?

iii) What is the analog signal $y_a(t)$ we can reconstruct from the samples if we use ideal interpolation?

Solution:

i) The frequency present in the signal above are $F_1 = 1kHz$, $F_2 = 3kHz$, and $F_3 = 6kHz$. Thus $F_{max} = 6kHz$ and the Nyquist rate is $F_N = 2F_{max} = 12kHz$.

ii) Since we have chosen $F_s = 5kHz$ and the folding frequency is ${}^{F_s}/_{2}$ = 2.5kHz

And this is the maximum frequency that can be represented uniquely by the sampled signal.

$$x(n) = x_a(nT) = x_a\left(\frac{n}{F_s}\right) = 3\cos 2\pi\left(\frac{1000}{5000}\right)n + 5\sin 2\pi\left(\frac{3000}{5000}\right)n + 10\cos 2\pi\left(\frac{6000}{5000}\right)n$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(\frac{3}{5}\right)n + 10\cos 2\pi\left(\frac{6}{5}\right)n$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(1-\frac{2}{5}\right)n + 10\cos 2\pi\left(1+\frac{1}{5}\right)n$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(-\frac{2}{5}\right)n + 10\cos 2\pi\left(\frac{1}{5}\right)n$$

$$= 13\cos 2\pi\left(\frac{1}{5}\right)n - 5\sin 2\pi\left(\frac{2}{5}\right)n$$

Among the frequency $F_1 = 1kHz$, $F_2 = 3kHz$, and $F_3 = 6kHz$. F_1 is less than folding frequency thus it is not affected by aliasing. However the other two frequencies are above the folding frequency and they will be changed by the aliasing effect.

iii) Since only the frequency components at 1kHz and 2kHz are present in the sampled signal, the analog signal we can recover is

$$y_a(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$$

Which is obviously different from the original signal $x_a(t)$. This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

3. A digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 3\cos 600\pi t + 2\cos 800\pi t$. The link is operated at 10,000 bits/sec and each input sample is quantized into 1024 different voltage levels.

i. What is the sampling frequency and folding frequency?

ii. What is the Nyquist rate for the signal $x_a(t)$?

iii. What is the frequencies in the resulting discrete time signal x(n)?

iv. What is the resolution Δ?

Solution:

i. Each input is quantized into L =1024 levels. The number of bits/sample are denoted by 'b' and given by $b = \log_2 L = \log_2 1024 = \frac{\log_{10} 1024}{\log_{10} 2} = 10 bits$

The rate in the digital communication link is given as Bit rate = samples/sec×bits/sample

Samples/sec is sampling frequency F_s = $\frac{\text{Bit rate}}{\text{bits/sample}}$= 1kHz

folding frequency = $\frac{F_s}{2} = 500Hz$

ii. The frequency present in the signal above are $F_1 = 300Hz$ and $F_2 = 400Hz$,

Thus $F_{max} = 400Hz$ and the Nyquist rate is $F_N = 2F_{max} = 800Hz$.

iii. The sampling frequency F_s = 1000Hz

The discrete time signal x(n) is given by putting t= nT =$\frac{n}{F_s}$

$$x(n) = 3\cos 2\pi \frac{300}{1000} n + 2\cos 2\pi \frac{400}{1000} n$$

$$= 3\cos 2\pi \frac{3}{10} n + 2\cos 2\pi \frac{4}{10} n$$

The frequencies in discrete time signal are $f_1 = 0.3\ and\ f_2 = 0.4$

iv. The two amplitudes are $A_1 = 3\ and\ A_2 = 2$ thus X_{max} = $A_1 + A_2$ = 5, X_{min} = $-X_{max}$= -5

$$\Delta = \frac{X_{max} - X_{min}}{L - 1}$$

$$\Delta = \frac{5 - (-5)}{1024 - 1} = 0.009775$$

Questions

1. Define energy and power signal with examples. Check whether the given signal is energy or power signal $x(t) = A\sin(t)$.
2. Define Signal Analysis and Signal Processing. What are the advantages of Digital signal processing over Analog signal processing?
3. Consider the analog signal $x_a(t) = 3\cos(100\pi t)$
   i. Determine the minimum sampling rate required to avoid aliasing.
   ii. What is the Nyquist rate for the signal $x_a(t)$?
   iii. If the signal is sampled at the rate $F_s = 200\ Hz$. What is the discrete-time signal obtain after sampling?
   iv. What is the frequency $0 < F < {}^{F_s}/_{2}$ of the a sinusoid that yields samples identical to those obtained for $F_s = 200\ Hz$.